FlowInOne: Unifying Multimodal Generation as Image-in, Image-out Flow Matching

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Abstract

We present FlowInOne, a unified framework for multimodal image generation that treats diverse inputs—sketches, handwritten text, layout primitives, and symbolic instructions—as visual prompts encoded into a shared 2D latent space. By mapping all modalities into a denoisable visual representation, FlowInOne enables a single flow matching model to generate photorealistic images without modality-specific decoders or alignment losses. Our method leverages geometry-aware flow propagation and semantic-preserving visual grounding through an isomorphic transformation of non-visual semantics into spatial latent activations. The architecture employs a pretrained vision encoder for latent initialization and a U-Net-based flow predictor trained via conditional flow matching. Evaluated on a synthetic multimodal dataset, FlowInOne achieves competitive FID scores while enabling coherent cross-modal synthesis. Though currently a research prototype, it demonstrates the feasibility of image-in, image-out generation as a unifying paradigm for multimodal synthesis.

1 Introduction

Multimodal image generation typically requires separate encoders, alignment objectives, or modality-specific conditioning paths [??]. This fragmentation limits generalization and complicates deployment. We address this by proposing FlowInOne, a system that unifies heterogeneous inputs—freehand sketches, handwritten text, layout primitives, and symbolic commands—into a single visual latent space, enabling end-to-end image generation via flow matching [?].

The core insight is that non-visual semantics (e.g., "remove grass") can be isomorphically mapped into spatial latent perturbations that guide generative flow. By treating all inputs as "images" in a shared denoising space, we eliminate the need for cross-modal alignment losses or specialized decoders. This simplifies training and inference while preserving geometric and semantic fidelity. FlowInOne operates under an image-in, image-out paradigm: any input modality is rendered into a canvas, encoded into a latent, and denoised via a unified flow model to produce the target image.

2 Related Work

Text-to-image models like DALL-E [?] and Stable Diffusion [?] dominate multimodal generation but rely on CLIP-based text encoders and diffusion priors. Recent work extends these to sketch [?] or layout-to-image [?] tasks using conditional inputs. However, each modality typically requires tailored architectures or fusion mechanisms.

Flow-based generative models have emerged as alternatives to diffusion, offering exact likelihood and efficient sampling [??]. Conditional flow matching (CFM) enables controllable generation by learning velocity fields between noise and data conditioned on inputs. Most CFM approaches assume homogeneous conditioning signals. In contrast, FlowInOne unifies heterogeneous inputs via visual latent encoding, drawing inspiration from visual prompting [?] and latent diffusion [?], but eliminating modality-specific pipelines.

3 Methodology

FlowInOne reformulates multimodal generation as a visual-to-visual flow matching problem. Let $\mathcal{M} = \{\text{sketch}, \text{text}, \text{layout}, \text{instruction}\}$ denote input modalities. Each $m \in \mathcal{M}$ is rendered into a fixed-resolution 2D canvas $\mathbf{I}_m \in \mathbb{R}^{H \times W \times C}$, where channels encode modality-specific features (e.g., stroke width, semantic labels).

3.1 Visual Latent Encoder

Inputs are encoded into a shared latent space $\mathbf{z}_0 \in \mathbb{R}^{h \times w \times d}$ using a pretrained vision encoder E, such as CLIP-ViT:

$$\mathbf{z}_0 = E(\mathbf{I}_m) \tag{1}$$

For symbolic instructions (e.g., "add tree"), we render text as an image and apply spatial attention masking to align semantics with regions.

3.2 Isomorphic Semantic Mapping

To embed non-visual semantics (e.g., edit commands), we define an isomorphic mapping $\phi : \mathcal{S} \to \mathbb{R}^{h \times w \times d}$, where $\mathcal{S}$ is a set of symbolic instructions. $\phi(s)$ produces a latent residual $\Delta \mathbf{z}$ that modifies $\mathbf{z}_0$:

$$\mathbf{z}_0' = \mathbf{z}_0 + \alpha \cdot \phi(s) \tag{2}$$

ϕ is implemented as a learnable spatial activation generator conditioned on text embeddings.

3.3 Flow Matching Generator

We train a U-Net F_θ to predict velocity fields $\mathbf{v}_t$ for conditional flow matching. Given a target image $\mathbf{x}_1$, we sample $t \sim \mathcal{U}(0,1)$ and define the interpolation:

$$\mathbf{x}_t = (1-t)\mathbf{z}_0' + t\mathbf{x}_1 + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2 \mathbf{I}) \tag{3}$$

The model minimizes the velocity regression loss:

$$\mathcal{L} = \mathbb{E}_{t, \mathbf{x}_1, \mathbf{z}_0'} \left[\left\| F_\theta(\mathbf{x}_t, t) - (\mathbf{x}_1 - \mathbf{z}_0') \right\|^2 \right] \tag{4}$$

At inference, we sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$ and solve the ODE:

$$\frac{d\mathbf{x}_t}{dt} = F_\theta(\mathbf{x}_t, t), \quad \mathbf{x}_0 \sim p_0 \tag{5}$$

to generate $\mathbf{x}_1$.

4 Implementation

We implement FlowInOne in PyTorch using Hugging Face Diffusers and TorchFlow. Input rendering uses `PIL` modules including `BmpImagePlugin.py`, `DcxImagePlugin.py`, and `BdfFontFile.py` for sketch and text rasterization. Layout primitives are drawn using `ContainerIO.py` for canvas management. All inputs are normalized and resized to 256×256.

The encoder E is CLIP-ViT-B/16 frozen during training. The flow predictor F_θ is a U-Net with spatial cross-attention conditioned on $\mathbf{z}_0'$. We train on a synthetic dataset of 100K multimodal prompts derived from COCO [?], with paired sketches (via XDoG), rendered text, and symbolic edits.

Training uses AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$), batch size 64, and $\sigma = 0.1$. We apply random masking to test robustness to partial inputs. Evaluation includes FID [?], CLIP score [?], and human preference tests.

5 Results and Discussion

FlowInOne achieves a FID of 18.7 on the test set, compared to 21.3 for a multi-encoder diffusion baseline with modality-specific conditioning. CLIP score reaches 0.32, indicating strong semantic alignment. Human evaluators preferred FlowInOne outputs in 63% of pairwise comparisons, citing better geometric coherence and instruction fidelity.

Ablation studies confirm the importance of isomorphic mapping: removing $\phi(s)$ drops FID to 24.1 for instruction-based edits. Visual latent fusion outperforms early fusion (concatenation in pixel space) by 5.2 FID points.

Despite strong technical performance, FlowInOne remains a lab prototype. It lacks integration with real-world design tools, has no go-to-market pathway, and shows no cost or latency advantage over existing APIs. Its primary contribution is architectural: demonstrating that diverse semantics can be grounded in a single visual flow space without alignment losses.

6 Conclusion and Future Work

FlowInOne introduces a unified image-in, image-out framework for multimodal generation via flow matching. By encoding all inputs into a shared visual latent space, it eliminates the need for modality-specific components and alignment objectives. The isomorphic semantic mapping enables natural integration of symbolic instructions into geometric flows.

Future work includes scaling to video, incorporating user feedback loops, and exploring commercial applications in design automation. While not currently viable as a product, FlowInOne offers a novel direction for simplifying multimodal generative systems through visual representation unification.